Document 2: The Technical Brief (The Pitch Benchmark)

Purpose: To prove your AI solves a multi-billion dollar bottleneck in the semiconductor industry.

OPTIQAI TECHNICAL BRIEF: FEBRUARY 2026
Subject: Accelerating the Electro-Industrial Stack via AI-Native Photonic Design

Founders: [Your Name], [CTO Name]

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1. THE PROBLEM: THE SIMULATION BOTTLENECK

In 2026, the scaling of AI data centers and custom ASICs is no longer limited by code, but by **interconnect bandwidth**. Silicon Photonics is the solution, yet the design process remains trapped in the 1990s. Traditional FDTD (Finite-Difference Time-Domain) solvers require hours of compute for a single iteration, creating a "Simulation Tax" that delays tapeouts by months.

2. THE OPTIQAI SOLUTION: NEURAL SURROGATE ENGINES

OptiqAI has developed **PhotonPilot**, an agent-native design environment powered by **Physics-Informed Neural Operators (PINO)**. We have successfully collapsed the loop between layout geometry and electromagnetic field validation from hours to milliseconds.

3. PERFORMANCE BENCHMARKS (V0.8 MVP)

We have benchmarked our AI Surrogate against the industry-standard Meep (3D-FDTD) on a 220nm SOI platform:

Component	Traditional FDTD (Meep)	OptiqAI Surrogate	Speedup
Ring Resonator	42 Minutes	**140 Milliseconds**	**~18,000x**
Grating Coupler	58 Minutes	**190 Milliseconds**	**~18,300x**
Accuracy (R^2)	1.0 (Ref)	**0.986**	High Fidelity

4. KEY INNOVATIONS

- **Physics-Aware Inference:** Unlike "black-box" AI, our models are constrained by Maxwell’s equations. The AI cannot "hallucinate" designs that violate the laws of physics.

- **Agent-Native Control Plane:** Integrated LLM agents that translate natural language (e.g., *"Optimize this MZI for C-band with <1dB loss"*) into verified GDSII layouts.
- **Cloud-Hybrid Sovereignty:** A containerized architecture allowing for VPC/On-Prem deployment—critical for protecting the proprietary IP of global semiconductor giants.

5. STRATEGIC POSITIONING

OptiqAI is positioned to be the "System of Record" for the global semiconductor shift:

- **USA:** Accessing risk capital to lead the AI-native industrial base.
- **Europe:** Leveraging the **EU Chips Act** and top-tier photonic R&D talent.
- **India:** Providing the design software stack for the emerging **India Semiconductor Mission (ISM)** fabs.

CEO Action Plan:

1. **Copy the text above** into a Google Doc.
2. **Save as OptiqAI_Technical_Brief_2026.pdf**.
3. **Visual Tip:** If you have a screenshot of your MVP's field prediction (the colorful "waves" of light), insert it right under the "Performance Benchmarks" table before saving. It makes the document look much more mature.